



Coding Thinker - Java Roadmap



Overview

A structured Java learning plan covering core language fundamentals and a focused DSA track. Each module lists estimated days and targeted practice questions.

Overview

Core Java Modules

DSA Track (with Java)

Arrays (15 days)

Strings (10 days)

Linked List (7 days)

Stack and Queue (15 days)

Recursion (7 days)

Backtracking (7 days)

Dynamic Programming (15 days)

Trees (7 days)

Heaps (Priority Queue) (5 days)

BST (Binary Search Tree) (5 days)

Graph (10 days)

Schedule and Tracking

▼ Core Java Modules

☐ 1) Java Intro, Syntax, Output, Comments · (2 days)

- Getting started, printing with `println` and `\n`, escape sequences
- Practice:
 - Create `public class Questions` and print "Hello"
 - Use `println`, `\n`, and escape sequences

☐ **2) Variables, Datatypes, Type Casting** · (3 days)

- Variables, naming, `final`, primitive vs reference types, casting
- Practice: final keyword, valid identifiers, datatype picks, widening vs narrowing casts

☐ **3) Operators (Complete)** · (4 days)

- Arithmetic, assignment, unary (inc/dec), relational, logical, ternary, bitwise
- Practice:
 - Percentage and total marks
 - Area of circle and 5 shapes
 - Swap numbers (normal and bitwise)
 - Even or odd, positive or negative
 - Reverse a number, leap year, Armstrong, palindrome
 - Divisibility checks
 - Interview-style logical and assignment operator questions

☐ **4) String, Math, Boolean** · (3 days)

- Strings: `==` vs `.equals`, String Pool, `StringBuffer` vs `StringBuilder`, common methods
- Math: Heron's formula (triangle area), quadratic roots

☐ **5) If-Else** · (2 days)

- Practice: circle area if radius > 0, grading, vowel check, toggle case, leap year

☐ **6) Switch** · (2 days)

- Practice: menu-driven calculator, multi-operation menu, weekday by number

☐ **7) While Loop** · (2 days)

☐ **8) For Loop, Break, Continue** · (2 days)

☐ **9) Loop Practice (Fundamentals)** · (2 days)

- 1 to 10, sum N, product N, multiplication table
 - Factorial, prime check, reverse number
 - Armstrong and palindrome via loops
 - Nth Fibonacci
- ☐ **10) Pattern Printing (Important)** · (2 days)
- Classic patterns with text, numbers, and stars including pyramids, inverted triangles, diamonds
 - Tip: derive loops using rows, columns, and spaces counters
- ☐ **11) Arrays** · (2 days)
- 1D and 2D arrays, traversal, aggregation, matrix operations
 - Practice: input and print, sum, element-wise add, 2D print, add two matrices, diagonals, transpose, multiplication
- ☐ **12) Methods** · (1 day)
- Defining and invoking methods, parameters, return values, overloading
- ☐ **13) Recursion (Basics)** · (1 day)
- Sum of N, range sum, factorial, Fibonacci
- ☐ **14) OOP (Classes, Objects, OOP Pillars)** · (6 days)
- Encapsulation, inheritance, polymorphism, abstraction
 - Constructors, `this`, `super`, access modifiers, static, `final`
 - Practice: interview-style OOP questions
- ☐ **15) Interface and Inner Classes** · (1 day)
- Interfaces, default/static methods, multiple inheritance via interfaces, inner/nested classes
- ☐ **16) Lambdas and Exceptions** · (1 day)
- Lambda basics, functional interfaces, try-catch, custom exceptions
- ☐ **17) Threads and File Handling** · (1 day)
- Thread lifecycle, synchronization concepts, basic file I/O
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DSA Track (with Java)

Arrays (15 days)

Includes 1D and 2D arrays, Kadane's algorithm, Dutch National Flag, searching, sorting, HashMap, HashSet, time and space complexity, sliding window, ArrayList.

- ☐ Largest element in array [GFG](#)
- ☐ Second largest (HW) [GFG](#)
- ☐ Third largest (HW) LC-414
- ☐ Reverse array, 2 methods [Naukri Code 360](#)
- ☐ LC-1752, LC-26 (HW), LC-283 (HW), LC-189, LC-485, LC-268, LC-136, LC-121, LC-75, LC-53, LC-48, LC-54, LC-1, LC-88, LC-169, LC-229 (HW), LC-73, LC-560
- ☐ Set union and intersection [GFG]
- ☐ Binary search: upper bound [GFG], lower bound (HW) [GFG], rotated array [GFG]
- ☐ LC-704, LC-33, LC-34, LC-81 (HW), LC-162, LC-540 (HW), LC-35 (HW)
- ☐ Sliding window: LC-3, LC-1004, Fruits into Baskets (HW) [GFG], plus the 4 standard variations
- ☐ Sorting practice: Bubble, Selection, Insertion

Strings (10 days)

Core string concepts covered in Core Java module.

- ☐ LC-28, LC-796 (HW), LC-1903, LC-242, LC-14 (HW), LC-151, LC-1021, LC-451

Linked List (7 days)

- ☐ Concepts and custom implementation
- ☐ CRUD operations
- ☐ LC-237, LC-234 (HW), LC-141, LC-876, LC-142, LC-2 (HW), LC-160 (HW), LC-2095 (HW), LC-206, LC-148, LC-61, LC-19 (HW)

Stack and Queue (15 days)

- ☐ Implementations and operations
- ☐ Prefix, infix, postfix conversions
- ☐ Min/Max stack
- ☐ LC-20, LC-735, LC-496 (HW), LC-402 (HW), LC-496, Next smaller/greater element, LC-503 (HW), LC-155 (HW), LC-239, LC-42 (HW)

Recursion (7 days)

- ☐ 10–15 foundational recursion problems
- ☐ Merge sort, quick sort, and related questions
- ☐ LC-8, LC-50 (HW), LC-1922 (HW), LC-39, LC-40, Subset Sum [GFG], permutations

Backtracking (7 days)

- ☐ Introduction and patterns
- ☐ N-Queens (LC-51), Rat in a Maze [GFG], LC-22, LC-78, LC-90 (HW), M-Coloring, Sudoku (HW)

Dynamic Programming (15 days)

- ☐ Memoization, tabulation, space optimization
- ☐ LC-70, Frog Jump (HW) [GFG], subsequences, House Robber (LC-198), LC-322, cut into segments, Ninja and the Fence, LC-62, LC-64 (HW), LC-377 (HW), LC-39, LC-40, LC-216, LC-300, LC-1143, LC-1027, LC-516, MCM

Trees (7 days)

- ☐ Intro, types, binary trees, traversals
- ☐ LC-144, LC-94 (HW), LC-145 (HW), LC-102, LC-100, LC-101 (HW), LC-110, LC-111 (HW), LC-543 (HW)
- ☐ Depth and diameter [GFG]

Heaps (Priority Queue) (5 days)

- ☐ Intro, heap operations, heap sort
- ☐ Priority Queue practice (LeetCode)

BST (Binary Search Tree) (5 days)

- ☐ Intro, implementation, properties
- ☐ LC-98, LC-96, LC-230, LC-700, LC-501, LC-235 (HW)

Graph (10 days)

- ☐ Intro, types, traversals
 - ☐ LC-1971, LC-997, LC-841
 - ☐ BFS core concept, LC-200, LC-785
 - ☐ DFS core concept, LC-695, LC-733, LC-130
 - ☐ Graph on grid / Matrix problems, LC-994, LC-542, LC-1091
 - ☐ Cycle detection & Bipartite, LC-207
 - ☐ Topological Sort, DAG, Khan's algo, BFS, DFS, LC-210
 - ☐ Shortest path algo, LC-787 LC-743
 - ☐ Minimum Spanning Tree(MST)
 - ☐ Union Find, LC-547, LC-684
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Schedule and Tracking

- Suggested pace: 3 classes per week, 1 hr 20 mins each ($\approx$ 4 hrs/week)
- Estimate for a combined Core + DSA journey depends on how many DSA problems you attempt weekly
- Tip: add checkboxes next to problems you solve and tag items with HW for homework